Single Image Haze Removal via Dark Channel Prior, with a Temporally Consistent Video Extension

Sohail Memon (2025201075) · Kathan Patel (2025201039) · Md Taufique Hussain (2025202007)

CS7.404 Digital Image Processing — Project Proposal

---

1 PROBLEM STATEMENT

Haze and fog degrade outdoor images by mixing true scene radiance with scattered ambient light (airlight) in proportion to scene depth, reducing contrast and shifting colour in a way that worsens with distance from the camera. This degradation is well described by the classical atmospheric scattering model

$$I(x) = J(x)\,t(x) + A(1 - t(x))$$

where I is the observed hazy image, J is the haze-free scene radiance we want to recover, A is the global atmospheric light, and $t(x) \in (0, 1]$ is the transmission map, which decreases with scene depth. Recovering J from a single image I is inherently under-constrained: three unknowns (J^r, J^g, J^b) plus t must be recovered per pixel from only three observed values, with A also unknown. He, Sun & Tang's **Dark Channel Prior (DCP)** (CVPR 2009, best paper award) closes this gap with a simple, empirically-grounded statistical observation about haze-free natural images, turning an ill-posed inverse problem into a solvable one without any learned component.

A second, related problem is that applying single-image dehazing independently to each frame of a video produces visible temporal flicker — the atmospheric light and transmission estimates jitter frame to frame even for a static scene, because both are derived from noise-sensitive extremal statistics (a single argmax pixel, and a local-minimum filter). This flicker is not addressed by the original DCP paper, which is single-image only.

2 PROPOSED SOLUTION

We will implement the Dark Channel Prior algorithm from first principles, and extend it with a temporally-regularized video pipeline evaluated as a controlled ablation.

Single-image pipeline:

1. **Dark channel computation** — a per-channel, per-patch minimum filter (grayscale morphological erosion), based on the empirical prior that in most local patches of a haze-free outdoor image, at least one colour channel has a pixel with very low intensity.
2. **Atmospheric light estimation** — selecting A from the brightest pixel(s) among the top 0.1% of pixels by dark-channel value.
3. **Raw transmission estimation** — derived directly from the dark channel of the (per-channel, A-normalized) hazy image.
4. **Transmission refinement — implemented two ways, compared directly:**

    - *Soft matting* (the paper's own method): solving a sparse linear system built from the Levin closed-form matting Laplacian, treating t as an alpha matte.
    - *Guided filter* (He, Sun & Tang's own faster follow-up, ECCV 2010): an edge-aware local linear filter with $O(N)$ cost, plus a fast/subsampled variant for near real-time throughput. This refinement-method ablation (quality vs. wall-clock cost) is a headline result in its own right.
5. **Scene radiance recovery** — direct inversion of the scattering model, with a lower bound on transmission to avoid noise amplification in high-depth regions.

Video extension (an ablation of temporal regularization strategies within a classical DCP pipeline): We diagnose the three sources of frame-to-frame flicker (atmospheric-light jitter, transmission jitter from the extremal min-filter, and boundary-clipping jitter), and build four pipeline variants of increasing sophistication:

- **V0** — per-frame DCP, no temporal handling (baseline).
- **V1** — exponential moving average (EMA) on atmospheric light only.
- **V2** — V1 + naive (unwarped) EMA on the transmission map.
- **V3** — V1 + optical-flow-warped EMA on the transmission map with a forward-backward occlusion mask, falling back to the raw per-frame estimate in occluded/moving regions to avoid ghosting.

No learned/deep-learning component is used anywhere in the dehazing method itself (a hard scope boundary); the pipeline remains entirely classical DIP.

3 OBJECTIVES

1. Implement the complete single-image DCP pipeline (dark channel, atmospheric light, raw transmission, recovery) and validate it against a synthetic round-trip test with known ground-truth transmission.
2. Implement and compare two transmission-refinement strategies — soft matting (the paper's own method) and guided filtering (a faster alternative), including a subsampled "fast guided filter" variant — quantifying both accuracy and runtime.
3. Benchmark the single-image pipeline on the RESIDE SOTS dataset (PSNR, SSIM, CIEDE2000) and situate results against published third-party DCP baselines.
4. Diagnose the specific causes of temporal flicker when DCP is applied frame-by-frame to video, with empirical evidence for which cause dominates.
5. Design, implement, and ablate four temporal-regularization variants (V0-V3) on the paired REVIDE video dataset, reporting per-frame fidelity (PSNR/SSIM) alongside temporal-consistency metrics (flow-warped flicker index, ROI variance) so the fidelity/stability trade-off is fully interpretable.
6. Characterize and document the method's known failure mode (sky and other intrinsically bright, colourless regions) with quantitative evidence, and evaluate one mitigation strategy.

4 DELIVERABLES

1. **Codebase** — a modular, config-driven implementation (`dcp/` for the single-image pipeline, `dcpvid/` for the video/temporal extension, `eval/` for metrics and benchmark runners), with every pipeline stage exposing its intermediate outputs (dark channel, A, raw/refined t, recovered J, depth map) for figures and debugging.
2. **Synthetic correctness test** — a round-trip unit test confirming recovered atmospheric light and transmission match known ground truth on synthetically hazed images.
3. **Single-image benchmark results** — PSNR/SSIM/CIEDE2000 and runtime on RESIDE SOTS (indoor + outdoor), with a one-factor-at-a-time ablation over refinement method, patch size, ω, atmospheric-light estimator, and t_0 .
4. **Video benchmark results** — per-frame PSNR/SSIM and temporal-consistency metrics (normalized flicker index, ROI variance, FPS) for all four temporal variants (V0-V3) on REVIDE, across a small sweep of the temporal smoothing coefficient.
5. **Failure-case analysis** — documented sky-region failure examples with a quantitative PSNR-vs-sky-fraction scatter, and one evaluated mitigation attempt.
6. **Before/after demonstration material** — side-by-side dehazed video clips (V0 vs. V3) and transmission-map visualizations for the presentation.
7. **Two project reports** (aligned to the course's Project Report 1 / Report 2 checkpoints) and a final presentation, plus a reproducibility-oriented README so every reported number can be regenerated from a single script.